Aqbulak Market: Modernizing the Business with Information Systems

SIS 3 Group Assignment, Fundamentals of Business for Information Systems

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Part A: Information Systems in Business

Information systems matter for a company working in several cities and countries because they give every store, warehouse and manager the same, current picture of the business. Without them, decisions rely on late and inconsistent reports. A shared system lets head office see stock, sales and supplier orders in real time across Almaty, Astana, Shymkent and, soon, Tashkent. With new regulations and currency in Uzbekistan, reliable information is essential for controlling costs and keeping customers satisfied.

Technical issue: Merging ten separate Excel files into one system, with inconsistent data formats and duplicate records.

Behavioral issue: Store staff used to their own spreadsheets may resist the new system or enter data carelessly.

Organizational issue: Each store works in its own way, so management must agree on common processes and who owns the data.

Part B: Enterprise Systems

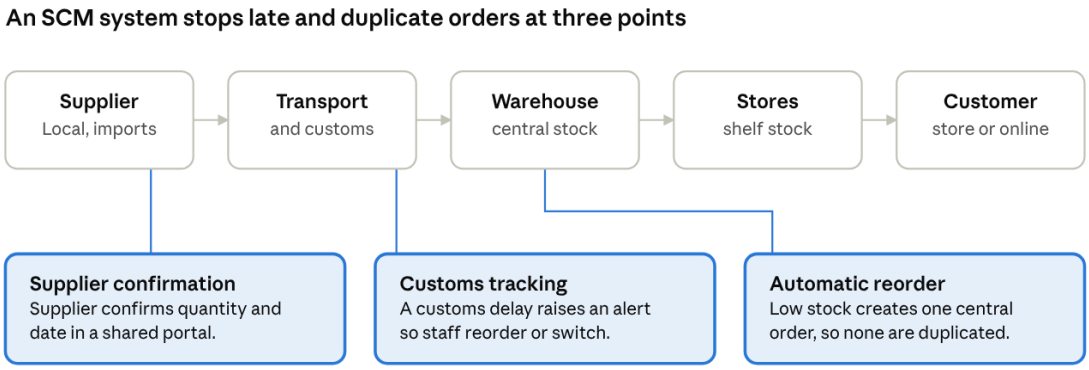
B1. ERP

An ERP (Enterprise Resource Planning) system is integrated software that runs core functions such as inventory, purchasing, sales and finance on one shared database. For Aqbulak, it replaces ten separate Excel files: when any store sells or receives goods, one central record updates instantly, so every manager sees the same stock levels.

- **Benefit 1:** Accurate real-time inventory reduces empty shelves and overstock.
- **Benefit 2:** Consolidated financial reports across all cities, including Tashkent, save time and reduce errors.
- **Challenge 1:** High cost of buying, customizing and training staff on the system.
- **Challenge 2:** Disruption during migration, since messy Excel data must be cleaned first.

B2. Supply Chain Management (SCM)

Simple supply chain:



Local producers deliver directly to the warehouse, while imports from Kyrgyzstan, China and Turkey first pass through transport and customs. An SCM system connects Aqbulak's stock data with suppliers. When stock falls below a set minimum, it proposes one purchase order and checks existing orders for the same product before approval, which prevents duplicates. Suppliers confirm quantities and delivery dates in a shared portal, so late orders are visible early. For imports, the system tracks shipments and customs documents and alerts staff to delays, so they can adjust plans or contact alternative suppliers.

B3. Customer Relationship Management (CRM)

A CRM would combine complaints and Aqbulak Card purchase history into one customer profile. Staff could see earlier issues and purchase preferences, with appropriate consent and access controls. For example, if a customer reports spoiled dairy, the store can resolve the complaint quickly and offer relevant loyalty points or a discount. Better service and personalized offers can encourage repeat visits.

Part C: E-commerce and Mobile Commerce

Business model

The most suitable e-commerce model is B2C (business-to-consumer), because Aqbulak sells groceries directly to individual shoppers. Click-and-collect is a fulfillment option within this model: customers order online and collect from a nearby store, or choose home delivery. Existing stores can serve as picking and dispatch points, limiting the need for separate warehouses.

Two supporting internet technologies are **cloud hosting**, which lets the website and app handle busy periods, and **SSL/TLS encryption**, which protects customers' payment and personal data.

Mobile commerce: three app features

With high smartphone use among shoppers, the app must make ordering and paying simple on a phone.

1. **Flexible payments:** Kaspi.kz, Kaspi Red/QR, Halyk Bank cards and cash on delivery. Customers pay the way they already do, so fewer abandon their baskets.

2. **Digital Aqbulak Card:** Points and personal offers appear in the app, so shoppers need no plastic card and have a reason to return.
3. **Delivery slots with order tracking:** Customers choose a time and receive push notifications, so they know when groceries will arrive.

Mobile risk: Payment fraud or stolen accounts on phones. This can be reduced with two-factor login and secure payment gateways.

Action plan to grow online sales

1. **Website and app:** Redesign the site mobile-first and launch the app with search, saved baskets and easy reordering.
2. **Payments:** Integrate Kaspi, Halyk and cash on delivery through a secure payment gateway.
3. **Delivery:** Pick orders in existing stores and use courier partners with fixed delivery slots, starting in Almaty and then Astana and Shymkent.
4. **Promotion and measurement:** Promote the app through Aqbulak Card offers, track completed and repeat orders, and set a proposed goal of raising online revenue share from 4% to 8% within 12 months.

Part D: Executive Memo

To: CEO, Aqbulak Market
From: Business Analyst Team
Subject: Which system to implement first

We recommend implementing ERP first because it replaces ten separate Excel files with one shared source of stock, purchasing and financial information. Managers will see available products across stores and make faster, more accurate decisions. Reliable inventory data will also support future SCM, CRM and e-commerce improvements, including expansion to Tashkent. The biggest risks are inaccurate data and staff resistance. We recommend cleaning and checking the existing spreadsheets, training employees, and testing ERP in two Almaty stores before expanding to the other locations. During the pilot, the team should monitor inventory accuracy and resolve problems before a full rollout. This gradual approach reduces disruption and makes the investment easier to manage.

Team contributions:

- Yessentur Nuraslanbek — Systems Analyst, Part A and B1 (ERP);
- Ashirbekov Adiz — Supply Chain and Customer Specialist, B2 (SCM) and B3 (CRM);
- Murat Alikhan — E-commerce Specialist, Part C;
- Bekmuratov Galymzhomart — Editor, consistency review and Part D;
- Bakir Didar — Presenter, group presentation preparation and Group Reflection.

Group Reflection

We learned that ERP provides shared data, SCM improves purchasing and deliveries, CRM supports customers, and e-commerce makes shopping convenient. We chose ERP first because the other systems depend on reliable information. Successful implementation also needs staff training, careful data migration and a gradual rollout. Next time, we would agree on shared terms and responsibilities earlier.